THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 1 ampere is equal to: 1

(a) 1 J/s **(b)** 1 J/C **(c)** 1 V/s **(d)** 1 C/s

2. Ammeter is connected in a circuit: 1

(a) across the cell **(b)** in parallel **(c)** in series **(d)** anywhere

3. Equivalent resistance of 10 Ω and 15 Ω in parallel is: 1

(a) 25 Ω **(b)** 5 Ω **(c)** 12.5 Ω **(d)** 6 Ω

4. A student states: “When resistors are connected in series, the current through each is the same.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only for equal resistors **(d)** Cannot be determined

5. Energy consumed by a 500 W heater in 4 hours is: 1

(a) 20 kWh **(b)** 2 kWh **(c)** 0.2 kWh **(d)** 2000 kWh

6. The direction of magnetic field at a point is given by the direction in which: 1

(a) the south pole of a compass points **(b)** iron filings fall
(c) the north pole of a compass points **(d)** the current flows

7. Magnetic field lines are close together near the: 1

(a) far away **(b)** poles
(c) middle of the magnet **(d)** nowhere

8. The magnetic effect of electric current was discovered by: 1

(a) Newton **(b)** Faraday **(c)** Ohm **(d)** Oersted

9. A student states: “Magnetic field lines can intersect at the poles.” The correct response is: 1

(a) Correct **(b)** Only inside
(c) Incorrect — they never intersect **(d)** Cannot be determined

10. The magnetic field due to a straight wire at a distance r is: 1

(a) directly proportional to r **(b)** inversely proportional to r
(c) independent of r **(d)** proportional to r^2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the resistance of a conductor? Write its SI unit. 2

12. Calculate the work done to move 5 C of charge across a potential difference of 6 V. 2

13. Draw (describe) the field lines around a straight wire carrying current upward. 2

14. Why does a current-carrying conductor experience a force in a magnetic field? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Two resistors of 3 Ω and 6 Ω are connected 3
(i) in series
(ii) in parallel to a 9 V battery. Find the current in each case.

16. How can you show that the magnetic field due to a solenoid is like that of a bar magnet? 3

17. State the factors on which the heat produced in a conductor depends. Write the formula. 3

OR

An electric motor takes 5 A from a 220 V line. Find its power and the energy consumed in 2 hours (in kWh and J). 3

18. State the factors affecting the magnetic field at the centre of a circular coil. 3

OR

What is meant by electric fuse? Explain its working in a domestic circuit. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State Ohm's law and its limitations. 5
(ii) Plot the V–I graph from the data V (V): 1.6, 3.4, 6.7, 10.2, 13.2; I (A): 0.1, 0.2, 0.4, 0.6, 0.8 (describe). Find R.

OR

Case study: In a house, a 2 kW geyser, three 100 W fans and five 20 W LED bulbs run on a 220 V supply. 5
(i) Find the total power.
(ii) Find the current drawn.
(iii) Which fuse rating is suitable: 5 A or 15 A?
(iv) Energy used if all run for 3 h.
(v) Why are they connected in parallel?

20. (i) What are magnetic field lines? 5
(ii) Describe an activity to draw them using a compass.
(iii) Write three properties.

OR

Case study: A wire carrying current from south to north is placed horizontally over a compass needle. 5
(i) In which direction does the north pole deflect?
(ii) What if the wire is below the needle?
(iii) Name the scientist who first observed this.
(iv) What does it prove?
(v) How can the deflection be increased?

END OF QUESTION PAPER — **Total: 40 Marks**